* Revision:

- data anay / data visu
- missing val
- outliers → ~~ ...

PHP
// myth //

splitting
↳ test

* Today's class:

① Two Fancy words:

a) Univariate analysis : Analysis single variable.

b) Multivariate analysis : Analyzing multiple variables together.



② Categorical Encoding techniques:

- What is Encoding ?

numerical

No. b No. bm am

nm

House price



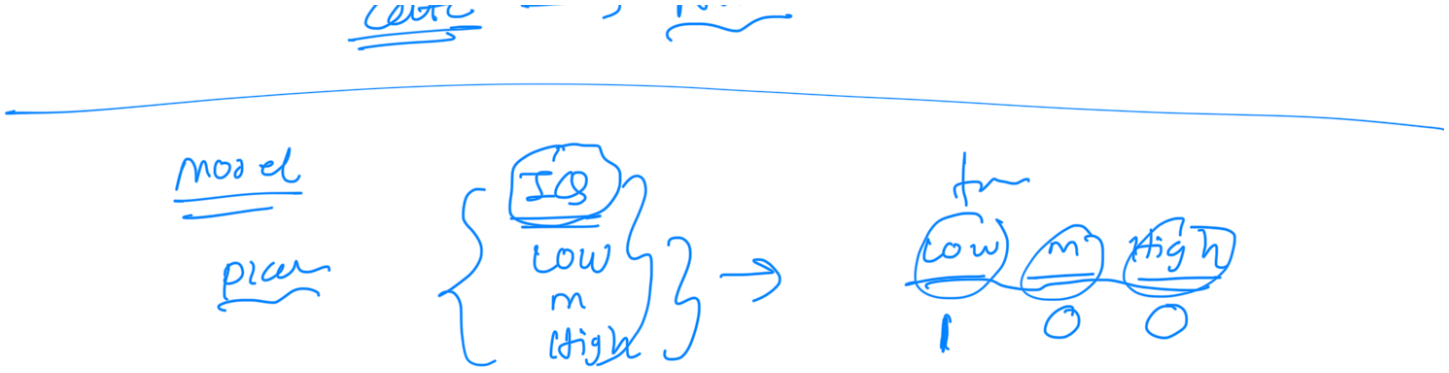
M2 model

MLR

$y = m_1 x_1 + m_2 x_2 + \cdots + c$

- Why Encoding is Require ?

cube → nm



@ One-hot Encoding:

- creates a seperate binary column for each category.

Ex:

color

↳ three cate: Red, Blue, Green → convert it into three column.

Single column

color → Red Blue Green

- data Example

color
Red
Red
blue
Green

Red	blue	Green
1	0	0
1	0	0
0	1	0
0	0	1

Pros

- No artificial order between categories.
- Easy to understand and implement.
- Works well with linear models, SVM, and many other algorithms.

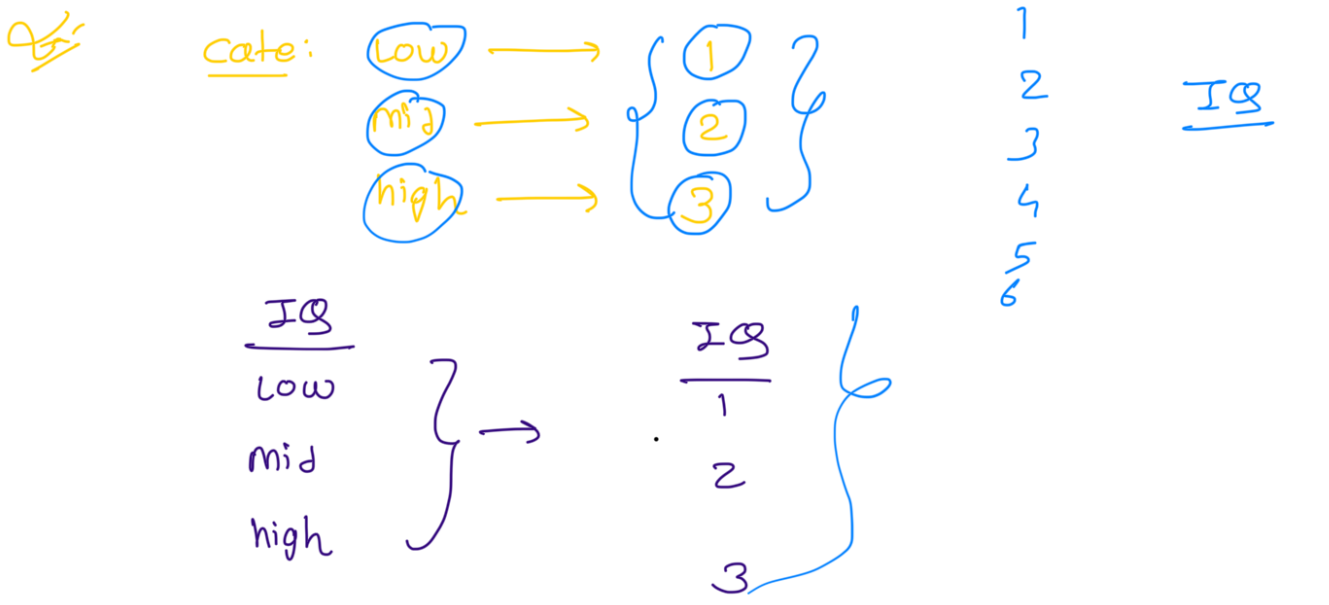
Cons

- Creates many columns when there are many unique categories.
- Can increase memory usage.
- Unseen categories need to be handled during inference.

Use when: The feature has a small or moderate number of nominal categories, such as gender, color, or payment method.

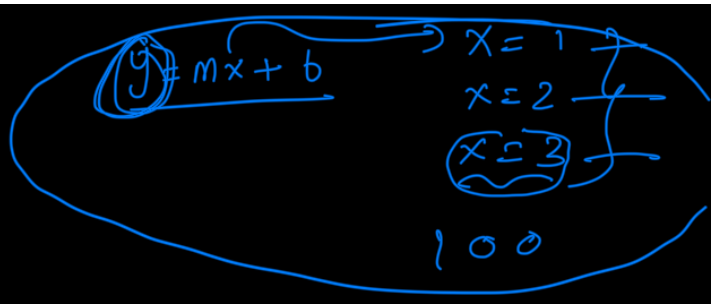
(b) Ordinal Encoding:

→ Assigns numbers according to a meaningful order.



Pros

- Simple and memory-efficient.
- Preserves the intended category order.
- Useful for genuinely ordered categories.



Cons

- The numeric distances may be misleading: the difference between 1 and 2 isn't necessarily equal to the difference between 3 and 4.
- Can mislead linear models if the numeric effect isn't appropriate.

Use when: Categories have a genuine order, such as low/medium/high, satisfaction ratings, or education levels.

ⓒ Label Encoding :

- assigns an integer to each category.

Chennai → 0
Delhi → 1
Ahem → 2
Noida → 3

Pros

- Simple and compact.
- Doesn't increase the number of columns.
- Can be useful with algorithms that explicitly support categorical features or for encoding target labels.

1 2 3 4

Cons

- Introduces an artificial numerical order for nominal categories.
- Can harm linear models, distance-based algorithms, and other models that interpret the values numerically.

Use when: Encoding the **target labels** in a classification problem, or when the model explicitly supports categorical values. For ordinary nominal input features, prefer one-hot encoding or another suitable method.

→ There are so many other techn.

② Feature Engineering :

- Feature Engineering is the process of creating, transforming or selecting input features to help a ML model learn useful pattern from data.

@ Creating New features :

Area (sq. ft.)	Bedrooms	Bathrooms	Price
1,000	2	1	₹30 lakh
1,500	3	2	₹45 lakh
2,000	4	3	₹60 lakh

Area

Price per bed

× 2

× 3

(b) Date & time :

month ~~date~~ day (1, 2, 3)

week_em (yes/no)

4th 3rd

month

year

Feature

Time

hour

am

mi

(c) Transformation ✓

(d) feature scaling & normalization ✓

(e) Encoding categorical features.

(f) Feature selection

(g) polynomial features

(h) Binning:

→ converts continuous variable into intervals.

Ex: Grading

Marks: go + : A

80 < m < go : B

70 < m < 80 : C

:

:

→ Feature Engi. is not only Applies to data we got.

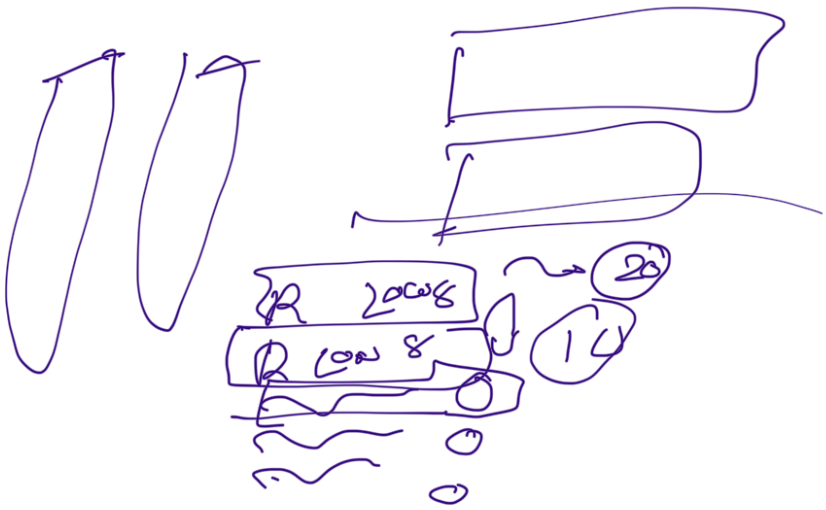
Netflix → Review system before

Netflix: → New Review system

like dislike

Simple UI change : 200% more Rating activity.

③ Duplicate values:

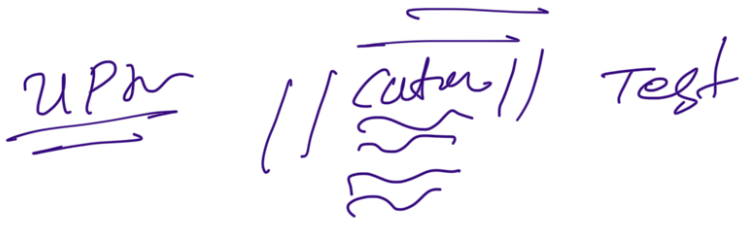


→ Pending:

- Scaling & Norm
- Transform
- Feature selection

→ 1) g.d 2) ML

(PCA) Regn / model based / statistical test


UPw
Custom
Test